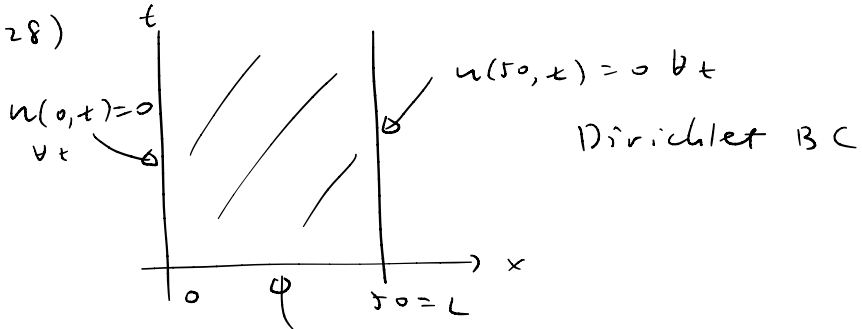


2025.12.11

Heat eqn: $u_t = \alpha^2 u_{xx}$

ex1 (p.628)



By the same procedure,

$$u(x, t) = \sum_{n=1}^{\infty} c_n e^{-\frac{n^2 \pi^2 \alpha^2}{50^2} t} \sin\left(\frac{n \pi x}{50}\right)$$

where

$$20 + f(x) = u(x, 0) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n \pi x}{50}\right)$$

$$\begin{aligned} \Rightarrow c_n &= \frac{2}{50} \int_0^{50} 20 \sin\left(\frac{n \pi x}{50}\right) dx \\ &= \frac{4}{5} \int_0^{50} \sin\left(\frac{n \pi x}{50}\right) dx \\ &= \frac{40}{n \pi} (1 - \cos(n \pi)) \quad \text{with } (-1)^n \text{ indicated} \\ &= \begin{cases} 0, & n \text{ even} \\ \frac{80}{n \pi}, & n \text{ odd} \end{cases} \end{aligned}$$

$$\Rightarrow u(x,t) = \frac{80}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} e^{-n^2 \pi^2 x^2 t / 50^2} \sin\left(\frac{n \pi x}{50}\right)$$

$$\text{or } u(x,t) = \frac{80}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} e^{-(2k+1)^2 \pi^2 x^2 t / 50^2} \sin\left(\frac{(2k+1)\pi x}{50}\right)$$

Neumann BC $\frac{\partial u}{\partial x}(0, t) = 0 \quad \forall t$

or

$$u_x(0, t) = 0$$

or both

$$u(0, t) + u_x(0, t) = 0 \quad \forall t$$

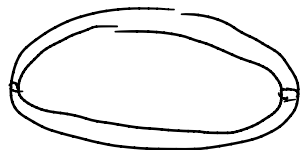
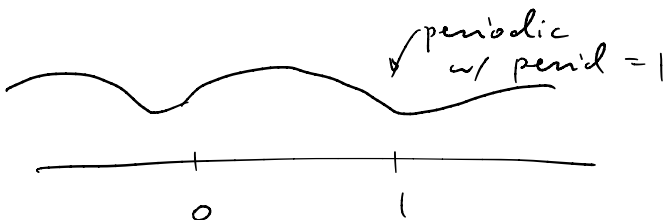
or $u(0, t) = -u_x(0, t) \quad \forall t$

or in general

$$\alpha_{11} u(0, t) + \alpha_{12} u_x(0, t) + \beta_{11} u(L, t) + \beta_{12} u_x(L, t) = 0$$

$$\alpha_{21} u(0, t) + \alpha_{22} u_x(0, t) + \beta_{21} u(L, t) + \beta_{22} u_x(L, t) = 0$$

⋮



$$u_t = \alpha^2 u_{xx} \quad \text{over } [0, 1]$$

length = 1

we identify points separated by distance

i.e., we identify the points $x=0$ & $x=1$

In terms of boundary condition, we have

$$u(0, t) = u(1, t) \quad \forall t$$

I.C

$$u_x(0, t) = u_x(1, t) \quad \forall t$$

$$u(x, 0) = f(x)$$

$$0 \leq x \leq 1$$

ϕ

$$\frac{\partial u}{\partial x}(0, t) = \frac{\partial u}{\partial x}(1, t)$$

periodic boundary condition.

Assume $u(x, t) = X(x)T(t)$

By separation variables, we get

$$\alpha^2 X'' T = X T' \Rightarrow \frac{X''}{X} = \frac{1}{\alpha^2} \frac{T'}{T} =: -\lambda$$

$$\Rightarrow \begin{cases} X'' + \lambda X = 0 \\ T' + \alpha^2 \lambda T = 0 \end{cases}$$

For the B.C.'s

$$u(0, t) = u(1, t) \Rightarrow X(0)T(t) = X(1)T(t) \quad \forall t > 0$$

$$\Rightarrow X(0) = X(1) \Rightarrow X(0) - X(1) = 0$$

$$u_x(0,t) = u_x(1,t) \Rightarrow X'(0)T(t) = X'(1)T(t) \quad \forall t > 0$$

$$\Rightarrow X'(0) = X'(1) \Rightarrow X'(0) - X'(1) = 0$$

Then the solution to $X'' + \lambda X = 0$ is

$$\textcircled{1} \lambda < 0, \quad X(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$$

$$X'(x) = c_1 \sqrt{-\lambda} e^{\sqrt{-\lambda}x} - \sqrt{-\lambda} c_2 e^{-\sqrt{-\lambda}x}$$

$$X(0) - X(1) = c_1 + c_2 - c_1 e^{\sqrt{-\lambda}} - c_2 e^{-\sqrt{-\lambda}} = 0$$

$$X'(0) - X'(1) = c_1 \sqrt{-\lambda} - c_2 \sqrt{-\lambda} - c_1 \sqrt{-\lambda} e^{\sqrt{-\lambda}} + c_2 \sqrt{-\lambda} e^{-\sqrt{-\lambda}} = 0$$

$$\Leftrightarrow \underbrace{\begin{bmatrix} e^{\sqrt{-\lambda}} - 1 & e^{-\sqrt{-\lambda}} - 1 \\ \sqrt{-\lambda}(e^{\sqrt{-\lambda}} - 1) & -\sqrt{-\lambda}(e^{-\sqrt{-\lambda}} - 1) \end{bmatrix}}_{A(\lambda)} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\det A(\lambda) = \underbrace{\sqrt{-\lambda}}_{\neq 0} \underbrace{(e^{\sqrt{-\lambda}} - 1)}_{\neq 0} \underbrace{(e^{-\sqrt{-\lambda}} - 1)}_{\neq 0} \underbrace{\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{\neq 0} \neq 0$$

$$\Rightarrow c_1 = c_2 = 0$$

$$\begin{bmatrix} \cancel{1} & \cancel{1} \\ \sqrt{\cancel{\lambda}} & \cancel{\lambda} \\ \ast & \ast \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ b \end{bmatrix}$$

$$\text{Try } \det(A^T(\lambda) A(\lambda)) = 0$$

$$\text{b/c } \ker(A) = \ker(A^T A)$$

$$\textcircled{2} \lambda = 0, \quad X(x) = c_1 x + c_2$$

$$X(0) - X(1) = c_2 - (c_1 + c_2) = 0 \Rightarrow c_1 = 0$$

$$X'(0) - X'(1) = c_1 - c_1 = 0$$

$$\text{so } X(x) = c_2, \quad c_2 \in \mathbb{R} \text{ const.}$$

$$\textcircled{3} \lambda > 0, \quad X(x) = c_1 \cos(\sqrt{\lambda} x) + c_2 \sin(\sqrt{\lambda} x)$$

$$X'(x) = -c_1 \sqrt{\lambda} \sin(\sqrt{\lambda} x) + c_2 \sqrt{\lambda} \cos(\sqrt{\lambda} x)$$

$$X(0) - X(1) = c_1 \cancel{\cos(\sqrt{\lambda} 0)}^1 + c_2 \cancel{\sin(\sqrt{\lambda} 0)}^0 - (c_1 \cos \sqrt{\lambda} + c_2 \sin \sqrt{\lambda}) = 0$$

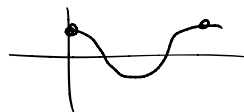
$$X'(0) - X'(1) = -c_1 \sqrt{\lambda} \cancel{\sin(\sqrt{\lambda} 0)}^0 + c_2 \sqrt{\lambda} \cancel{\cos(\sqrt{\lambda} 0)}^1 - (-c_1 \sqrt{\lambda} \sin \sqrt{\lambda} + c_2 \sqrt{\lambda} \cos \sqrt{\lambda}) =$$

$$\Rightarrow \underbrace{\begin{bmatrix} \cos \sqrt{\lambda} - 1 & \sin \sqrt{\lambda} \\ -\sqrt{\lambda} \sin \sqrt{\lambda} & \sqrt{\lambda} (\cos \sqrt{\lambda} - 1) \end{bmatrix}}_{A(\lambda)} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (*)$$

$$\det A(\lambda) = \sqrt{\lambda} (\cos \sqrt{\lambda} - 1)^2 + \sqrt{\lambda} \sin^2 \sqrt{\lambda} = 0$$

$$\Rightarrow \underbrace{\sqrt{\lambda}}_{\neq 0} (2 - 2 \cos \sqrt{\lambda}) = 0$$

$$\Rightarrow \cos \sqrt{\lambda} = 1$$



$$\Rightarrow \sqrt{\lambda} = 2\pi n, \quad n = 1, 2, 3, \dots$$

$$\Rightarrow \lambda = (2\pi n)^2, \quad n = 1, 2, 3, \dots$$

when $\lambda = (2\pi n)^2$, $n = 1, 2, 3, \dots$

(*) becomes

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow c_1, c_2 \in \mathbb{R}$ are arbitrary const.

$$\Rightarrow X(x) = c_1 \cos(2\pi n x) + c_2 \sin(2\pi n x)$$

~/ $c_1, c_2 \in \mathbb{R}$ are arbitrary const.

when $\lambda = 0$, $T' = 0 \Rightarrow T(t) = c = \text{const.}$

$$\text{take } u_0(x, t) = 1$$

$$\text{when } \lambda > 0, \quad T' + \alpha^2 (2\pi n)^2 T = 0$$

$$\Rightarrow T(t) = T(0) e^{-4\pi^2 \alpha^2 n^2 t}$$

we take fundamental solutions as

$$u_n^{(1)}(x, t) = e^{-4\pi^2 \alpha^2 n^2 t} \cos(2\pi n x) \quad n = 1, 2, \dots$$

$$u_n^{(2)}(x, t) = e^{-4\pi^2 \alpha^2 n^2 t} \sin(2\pi n x) \quad n = 1, 2, \dots$$

so the general solution is given by

$$u(x, t) = \frac{a_0}{2} + \sum_{n \geq 1} e^{-4\pi^2 \alpha^2 n^2 t} (a_n \cos(2\pi n x) + b_n \sin(2\pi n x))$$

Set $t = 0$, then $\frac{1}{T} \int_0^T f(x) dx \quad (T=1)$

$$f(x) = u(x, 0) = \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos(2\pi n x) + b_n \sin(2\pi n x))$$

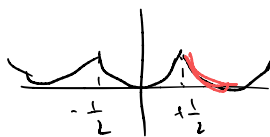
(Again multiply $\cos(2\pi nx)$ or $\sin(2\pi nx)$ then integrate over $[0, 1]$)

$$a_n = \frac{\int_0^1 f(x) \cos(2\pi nx) dx}{\int_0^1 \cos(2\pi nx)^2 dx = \frac{1}{2}} = 2 \int_0^1 f(x) \cos(2\pi nx) dx$$

$n = 0, 1, 2, \dots$

$$b_n = 2 \int_0^1 f(x) \sin(2\pi nx) dx, \quad n = 1, 2, 3, \dots$$

ex $f(x) = x^2 \quad -\frac{1}{2} \leq x \leq \frac{1}{2}$



$$a_n = 2 \int_{-1/2}^{1/2} x^2 \cos(2\pi nx) dx$$

$$= \frac{(-1)^n}{n^2 \pi^2} \quad n \geq 1$$

$$\frac{a_0}{2} = \int_{-1/2}^{1/2} x^2 dx = \frac{1}{12}$$

$$b_n = 0 \quad \forall n$$

$$\begin{aligned} & \int_{-1/2}^{1/2} x^2 \cos(2\pi nx) dx \\ &= \int_0^1 (x-1)^2 \cos(2\pi nx) dx \end{aligned}$$